RR

- 01 Executive Overview
- 02 Spatial Inequality
- 03 Road Risk & Severity
- 04 Data Quality

ROAD SAFETY INTELLIGENCE

Road Collision Risk & Spatial Inequality

Where is road risk high after exposure, and where is collision burden socially concentrated?

↻ Reset filters

YEAR: All

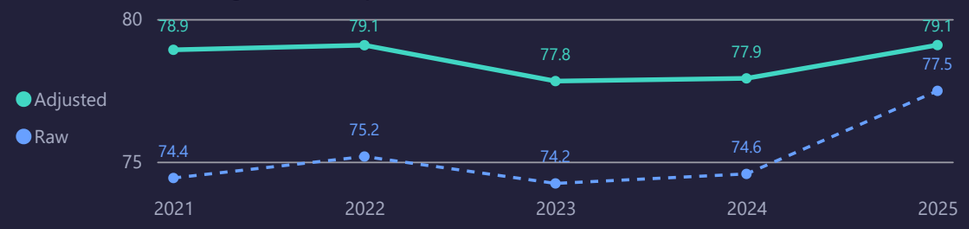
RISK METRIC: Adjusted expected KSI

ⓘ Resident-normalised rates show community burden, not traffic exposure.

Reported collisions	Adjusted KSI / bn vehicle miles	Deprivation burden ratio	Traffic-exposure coverage
513,801	78.6	2.03x	99.98%

NATIONAL TREND

KSI risk after accounting for traffic exposure



DECISION SIGNAL

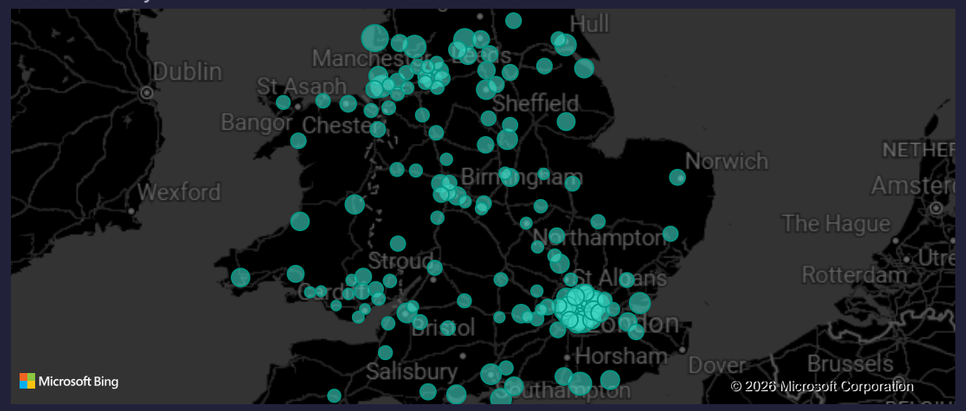
Two problems require two intervention tracks

2.03× Five-year collision burden: most deprived versus least deprived neighbourhoods.

30.9 Adjusted expected KSI per 100 rural collisions in the selected period.

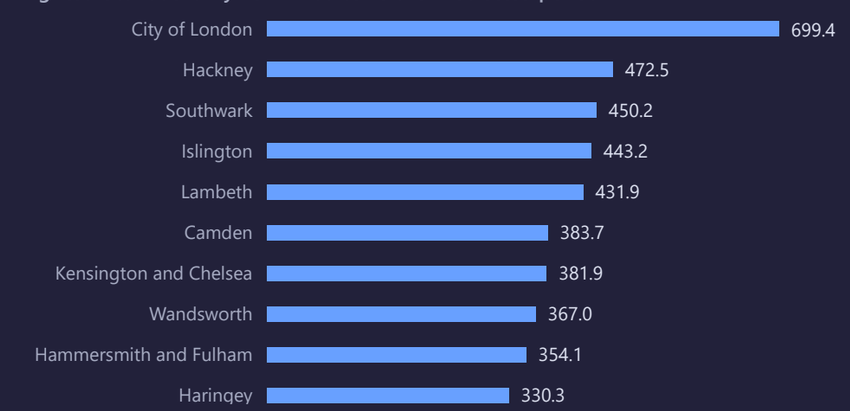
GEOGRAPHIC DISTRIBUTION

Local-authority KSI risk · Bubble size = selected metric



INVESTIGATION QUEUE

Highest local-authority risk · ≥100 collisions in selected period



● Reviewed data
2021 -2025

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ROAD SAFETY INTELLIGENCE

Road Collision Risk & Spatial Inequality

Separate community burden from traffic exposure before setting place-based priorities.

↻ Reset filters

YEAR

All

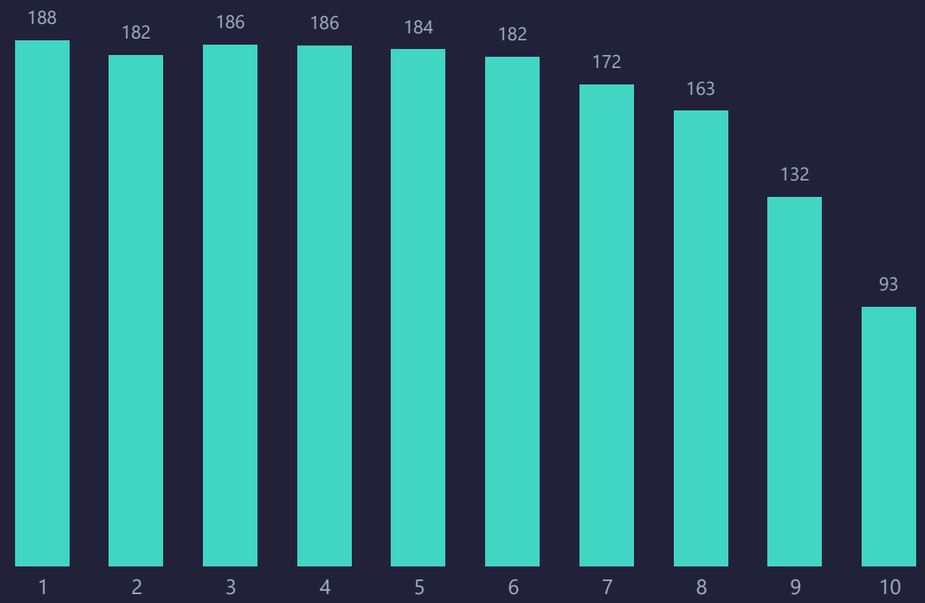
SPATIAL INEQUALITY

The deprivation gap reflects collision occurrence rather than conditional severity.

ⓘ Resident-normalised rates show community burden, not traffic exposure.

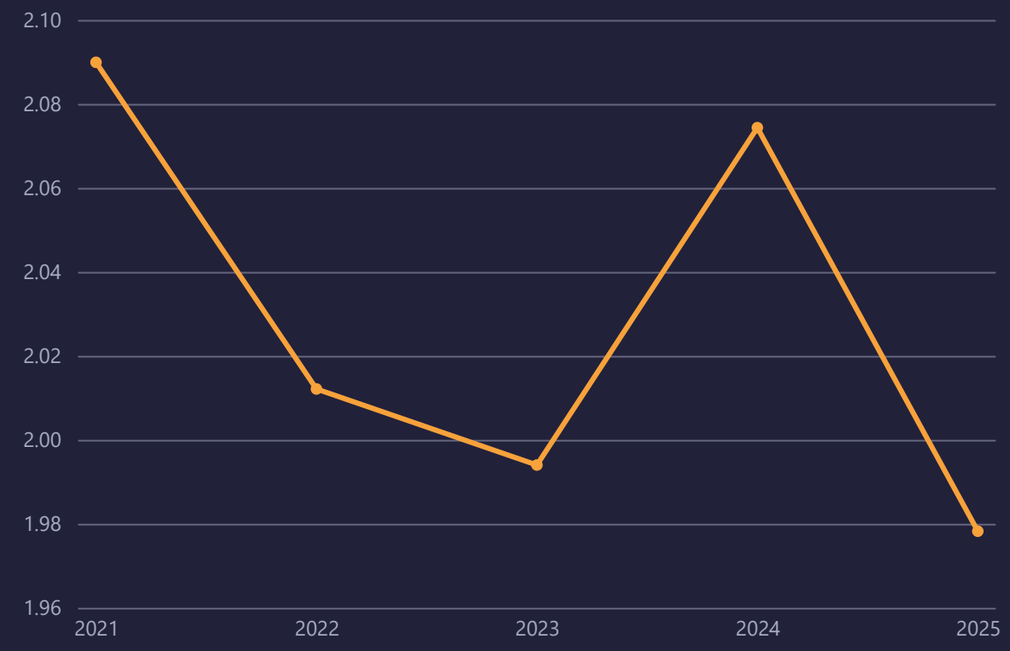
Collision burden by deprivation decile

Collisions per 100k residents per year



Annual most-to-least deprived gap

Zoomed scale · 1.00 = parity



INTERPRETATION

The inequality signal is persistent, but it is not an exposure-adjusted road-risk estimate.

Use this view to prioritise neighbourhood diagnosis and place-based prevention. Do not use it to claim that deprivation causes collisions or that residents themselves generated the traffic exposure.

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ROAD SAFETY INTELLIGENCE

Road Collision Risk & Spatial Inequality

Compare collision consequences across road environments without implying causality.

↻ Reset filters

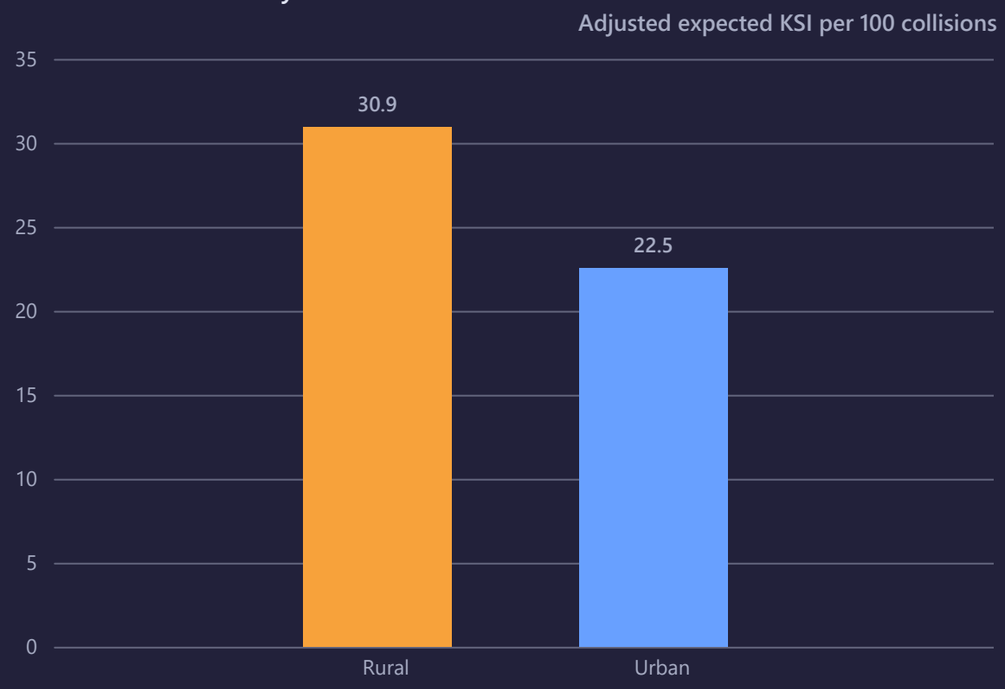
YEAR

All

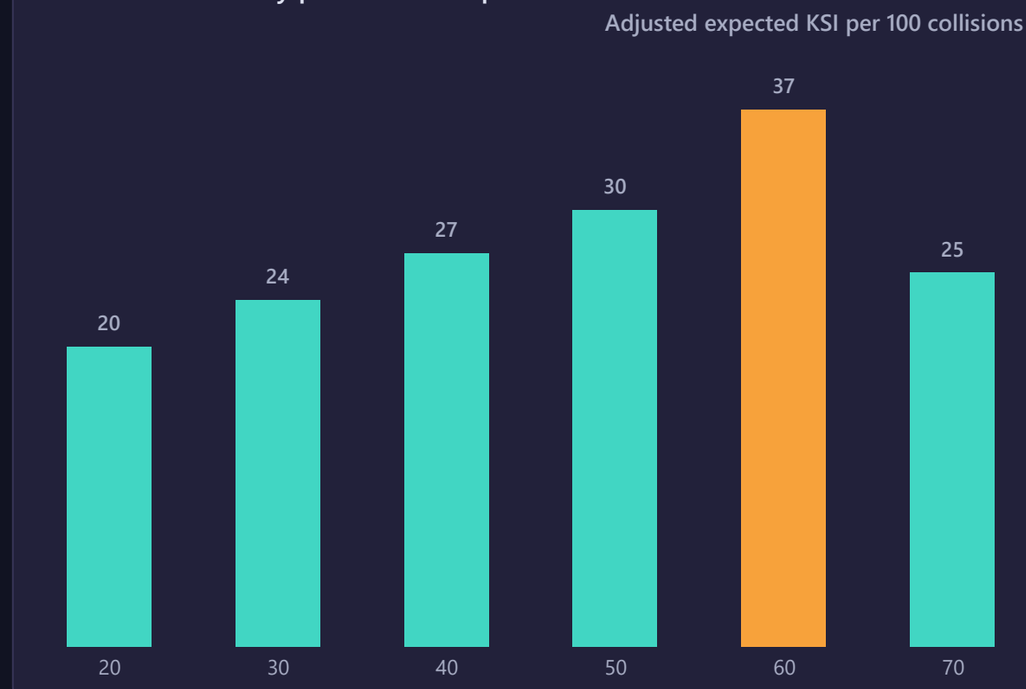
ROAD RISK & SEVERITY

Context changes the consequences of a collision

Urban vs rural severity



Conditional severity peaks on 60 mph roads



INTERPRETATION

Speed limit is informative, but not a sufficient causal explanation.

Conditional severity peaks on 60 mph roads and falls on 70 mph roads. Road class, traffic separation, junction context and road-user mix should be modelled jointly before recommending a specific intervention.

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2021 -2025

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ROAD SAFETY INTELLIGENCE

Road Collision Risk & Spatial Inequality

Exposure coverage, matching limits and measurement sensitivity beside the analysis.

↻ Reset filters

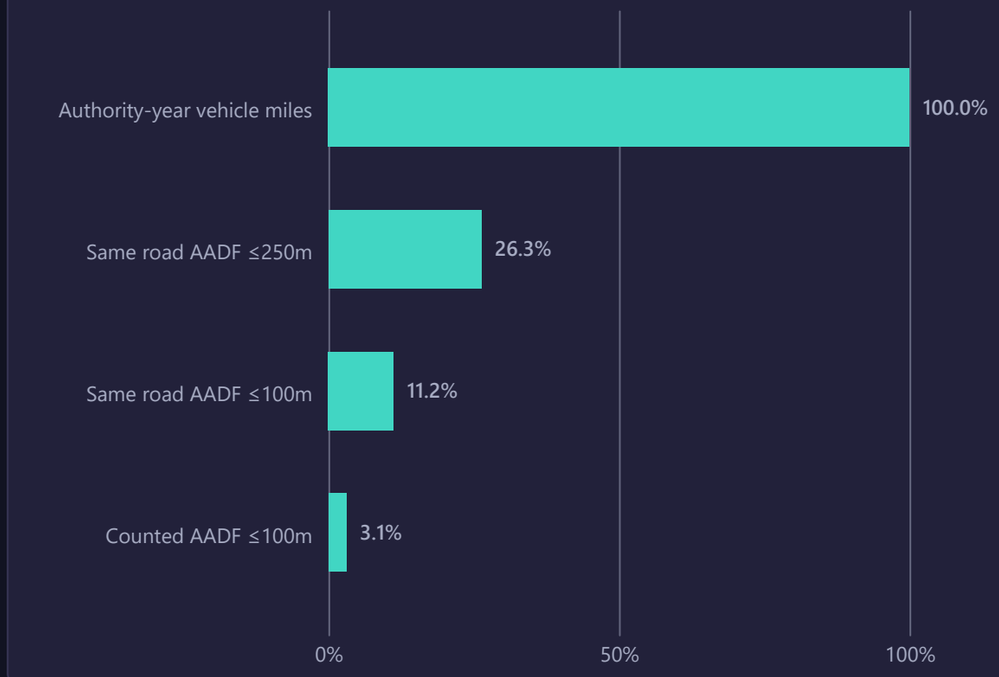
YEAR

All

DATA QUALITY & SENSITIVITY

Make analytical confidence visible beside the result

Exposure scenario coverage



MODEL GUARDRAILS

Local-authority traffic coverage

99.98%

Authority-year vehicle miles

England-to-IoD join

100%

Applicable English collision records

AADF ≤100m coverage

11.2%

Eligible numbered-road collisions

Raw vs adjusted rank correlation

0.991

2025 Spearman ρ

LIMITATION

AADF point matches are a sensitivity diagnostic, not a link-level absolute risk denominator.

Do not use the AADF coverage scenarios to publish absolute road-link risk rates until road-segment geometry matching is available.

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